1. MultiPhysio-HRC: A Multimodal Physiological Signals Dataset for Industrial Human–Robot Collaboration

Methodology

Created a Rich Multimodal Dataset

They collected a new dataset called **MultiPhysio-HRC**, which includes synchronized human physiological and behavioral signals recorded during real and realistic human–robot collaboration scenarios.

The **modalities included**:

- **Electroencephalography (EEG) –** brain electrical activity
- **Electrocardiography (ECG)** – heart activity
- **Electrodermal activity (EDA) –** skin conductance (stress indicator)
- **Respiration (RESP)**
- **Electromyography (EMG)** – muscle activity
- **Voice recordings**
- **Facial action units (facial expression features)**
  These modalities together offer a *comprehensive view of mental, emotional, and physical states*.

Collected Data in Diverse Tasks

The dataset was not limited to one type of task. It includes:

Controlled Cognitive Tasks

Designed to elicit different levels of:

- **Stress**
- **Cognitive load**
  Examples include Stroop tasks, n-back working memory tests, arithmetic tasks, and problem-solving tasks.

Immersive VR Stress Scenarios

Virtual reality experiences *designed to induce stress* and emotional responses.

Industrial-Like Collaboration with a Robot

Participants performed:

- Manual industrial tasks (e.g., battery disassembly)

- Robot-assisted tasks with a collaborative robot
  This places data collection in *a realistic human–robot collaborative context*, not just in lab conditions.

Baseline Models & Evaluation

The authors evaluated baseline machine learning models on the dataset to:

- Predict stress
- Predict cognitive load
  They showed that these physiological and multimodal signals *do contain meaningful information about human mental states*, but also highlighted the challenges in achieving high accuracy — pointing to the need for more advanced models and multimodal fusion techniques in future research.

Using MultiPhysio-HRC in Cobot Projects:

Human Affective & Cognitive State Awareness

Cobots traditionally respond to **physical motion or proximity** (e.g., force, gestures, speed).
This dataset enables robots to understand **internal states** such as:

- Cognitive load (workload)
- Stress and anxiety
- Physiological arousal
- Muscle fatigue (via EMG)
- Emotional valence

This can be critical for **adaptive behavior** — e.g., a robot slowing down or offering more assistance when the human is stressed or overloaded.

Example Applications:

- Safety systems that reduce robot speed during high human stress
- Workload-aware task allocation
- Adaptive assistance levels in collaborative assembly

Multimodal Sensor Fusion for Better Human Perception

The dataset includes synchronized multimodal signals such as:

- **EEG** — brain activity
- **ECG** — heart rate dynamics
- **EDA** — skin conductance

- **RESP —** respiration patterns
- **EMG —** muscle activity
- **Facial features**
- **Voice recordings**

This rich sensory input is ideal for research into:

- Multimodal adaptation models
- Deep learning fusion algorithms
- Real-time intent/state estimation

Such models can make a cobot **more perceptive and responsive** than if it only used vision or motion data.

Affective Human–Robot Interaction (HRI)

Understanding emotional and cognitive human states enables:

- Robots that *empathize* or *adjust behavior* based on human stress
- More natural and safe interactions
- Reduction of human workload and errors

For example:

- Robot reduces speed if human is stressed
- Robot offers suggestions or pauses when workload is high
- Robot adjusts communication style based on affect

2. HARMONIC: A Multimodal Dataset of Assistive Human–Robot Collaboration

Methodology

1) Designed a Real Assistive HRC Task

Participants collaborated with a 6-DoF robot arm (e.g., platforms from KUKA) to perform an **assistive feeding/manipulation task**.
The human provided high-level intent and corrective inputs; the robot executed low-level motion—typical **shared autonomy**.

2) Collected Synchronized Multimodal Data

They recorded tightly time-aligned streams from **human, robot, and environment**:

- **Human sensing**
  - Eye gaze (binocular eye cameras)
  - Egocentric (first-person) RGB video
  - Forearm EMG (muscle activity)
  - Joystick inputs (control intent)
- **Robot sensing**
  - Joint positions/kinematics
  - End-effector state
- **Third-person perception**
  - Stereo video of the scene

They also provide **processed annotations/features**, such as:

- Gaze projected into egocentric video
- 2D/3D body and hand pose
- Facial keypoints

All modalities are **time-synchronized**, enabling causal modeling across signals.

3) Structured the Dataset for Learning Problems

They organized trials into task phases (approach, manipulation, correction, completion) so the dataset can support:

- **Goal/intent prediction** (early inference of user goal)

- **Action anticipation** (predict next robot motion from human signals)
- **Shared autonomy policies** (assistive control blending)
- **Multimodal fusion** (vision + gaze + EMG + control)

4) Baseline Benchmarks

The authors report baseline results for representative tasks (e.g., intent inference and action prediction) to show:

- Multimodal models outperform single-modality baselines
- Gaze and control inputs are strong early indicators of intent
- Physiological/control cues add predictive value beyond vision alone

This establishes **benchmark protocols** for future methods.

Core Contributions

- **A high-quality, synchronized multimodal HRC dataset** (human perception + physiology + control + robot state).
- **Real shared-autonomy interaction**, not passive observation or teleoperation logs alone.
- **Standardized benchmarks** for intent prediction and assistive control learning.
- **Tooling and annotations** that reduce friction for multimodal learning research.

What others did with this dataset (domain, year of release, ...)

Where HARMONIC Fits in a Cobot System

1) Shared Autonomy & Intention Inference

HARMONIC includes **human control inputs (joystick), gaze, EMG, egocentric/third-person vision, and robot kinematics** captured during real collaborative assistance. This supports:

- Early **goal/intent prediction**
- **Action anticipation** (predict next robot motion from human signals)
- **Control blending** policies for shared autonomy

test algorithms

These are core problems in cobot interaction.

2) Multimodal Human State Estimation

Because modalities are synchronized, you can study:

- **Multimodal fusion** (vision + gaze + EMG + control)
- **Robust perception under partial observability**

- **User adaptation / personalization** across subjects

This is directly applicable to adaptive cobots that tailor assistance to the operator.

3. A Dataset of Standard and Abrupt Industrial Gestures Recorded Through MIMUs

Methodology

1. Created the DASIG dataset
They designed and collected a dataset called **DASIG (Dataset of Standard and Abrupt Industrial Gestures)** using MIMUs (Magneto-Inertial Measurement Units — wearable motion sensors).
How to measure safety using this dataset, test algorithms

2. Participants & tasks

- **60 healthy adult participants** took part.
- Participants performed a sequence of:
  - *Standard industrial gestures* (e.g., typical pick-and-place tasks).
  - *Abrupt movements* triggered by stimuli such as **visual alarms (lights)** or **auditory alarms (buzzers)** to simulate unexpected situations.

3. What the dataset includes

- Sensor signals from multiple MIMUs recorded during movement.
- Temporal information about when alarms were triggered.
- Anthropometric (body measurement) data for each participant.
- Example code/scripts showing how to use the dataset.

Why It Matters

- **Rich data for ML and robotics**: This dataset can be used to train and test algorithms for gesture recognition, safety systems, and human-motion-aware robot control.
- **Focus on abrupt motion**: The inclusion of sudden unplanned movements helps develop more *robust and safe* industrial robots that can react properly when humans behave unpredictably.
- **Applicable for ergonomics and safety research**: Besides robotics, researchers in ergonomics, motion planning, and human–machine interaction can also use this dataset.

Why It *Is* Useful for Collaborative Robots (Cobots)

The dataset contains:

- Wearable **MIMU (IMU + magnetometer) signals**

- Standard industrial gestures
- Abrupt/sudden movements triggered by alarms
- Multiple subjects (inter-subject variability)
- Temporal annotations

This makes it valuable for:

1. Human Motion Recognition

You can train models to:

- Recognize normal work gestures
- Detect abnormal or abrupt movements
- Classify motion phases

This is directly useful in:

- Human intention recognition
- Gesture-based robot control
- Adaptive robot behavior

2. Safety in Human–Robot Collaboration

Abrupt movements are extremely important in HRC.

Example uses:

- Detecting panic or emergency reactions
- Detecting sudden human withdrawal
- Predicting collision risk
- Triggering robot slowdown or stop

This aligns very well with **ISO safety requirements for collaborative robots**.
Respect ISO normes